Assignment: Deep Learning Optimization (Computer Vision)

Topic: Manual Weight Update using the Adam Optimizer

Course: Computer Vision

Status: Practice Exercise N2

Sessional Marks : 05

The objective of this assignment is to understand the mechanics of the **Adaptive Moment Estimation (Adam)** optimizer. You are required to perform a manual calculation of a weight update for a single step (t=1) given a specific set of hyperparameters and initial conditions.

Problem Statement

Given a single weight in a model, calculate the **updated weight** (w1) after one step of the Adam optimizer using the values provided below.

Given Parameters:

Initial Weight $(\omega_0): 1.20$

$Gradient(g_1): 0.15$

$LearningRate(\eta): 0.05$

$First\ Moment\ Decay(\beta_1): 0.9$

$Second\ Moment\ Decay(\beta_2): 0.99$

$Epsilon(\epsilon$

Initial Moments $(m_0,\ v_0): 0$

Deadline

Submission Date: April 19, 2026

Day: Sunday

Note: Submissions made after the deadline may result in a deduction of marks.